



Strengthening Drought Monitoring Module by Ensembling Auxiliary Information Based Varying Estimators

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Abstract

Quality of meteorological data such as preciseness and accuracy has substantial importance while making inferences about natural hazards. Among all the natural hazards, drought hazard is a complex natural phenomenon. In this article, we have discussed the procedure of improving time series data of meteorological indicators for analyzing drought. Here, we considered auxiliary information-based sampling estimators to enhance the quality of drought indicators. Since precipitation time series has a key role in drought occurrence and have a strong spatial correlation coherence structure with temperature. Therefore, this study suggests the use of the characteristics of auxiliary information as local weights for improving precipitation records. Consequently, this study presents the proposal of a new drought measure: The Seasonally Transient Weighted Multi-Scaler Standardized Index (STWMSDI). Under varying relationships between time series data of precipitation and temperature, the proposed method is more general than presented in Ali et al. (2019). We applied STWMSDI on ten meteorological stations of Pakistan to compare its performance with the Standardized Precipitation Index (SPI). Experimental results show high significant correlation between the STWMSDI and SPI. From these results, we conclude that the improved data provides much better results in terms of drought indices. Hence, the STWMSDI method of drought index is a good candidate for accurate drought monitoring.

Keywords Drought · Auxiliary information · Sampling estimator · Drought indices

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1 Introduction

Drought – an extended period of water deficit at a specific region (Dracup et al. 1980), is a major stochastic natural hazards (Hao et al. 2018; Sutanto et al. 2020). A drought is an extreme event that harms the natural environment and human lives (Mishra and Singh 2010). Its impact remains prolonged for several years. Drought is usually characterized by its properties (i.e. severity, duration, and real extent) Tigkas et al. (2015).

In the monetary term, drought costs 21.6 billion dollars in 2012 in the United States that is collectively more than any other natural disaster (Nielsen-Gammon 2012; Seager et al. 2015; Elliott et al. 2018). In recent decades, drought has caused remarkable losses in agriculture yield (Dutta et al. 2013; Hoerling et al. 2014; Ortiz-Bobea et al. 2019), water supply (Frick et al. 1990), and ecosystem (Lake 2003) in several regions. Therefore, its accurate monitoring is essential for drought mitigation policies.

A histrionic consequence of climate change is the alteration of hydro-meteorological cycles, with increasing intensity and frequency of extreme events such as droughts Wilhite (1993, 2000). Numerous studies have evidence that the pattern of global climate is rapidly changing (Kornhuber et al. 2018). Consequently, significant changes have been observed in the summary statistics of stochastic variables related to atmospheric and geosciences. Comparative to other, significant variations in the distribution of rainfall and temperature have been reported (Khozeymehnezhad et al. 2019). Climatic variables such as precipitation, humidity, evapotranspiration, and temperature play a key role in the occurrence of drought, however, the distribution of precipitation is the most reliable variable in drought occurrence and existence (Apurv et al. 2017). Furthermore, the amount and distribution of precipitation have a significant effect on hydrological and meteorological cycles. The amount and distribution of precipitation are the main factors in the whole ecosystems (Aswathanarayana 2001).

In literature, many authors have suggested various procedures for drought monitoring (McKee et al. 1993; Tsakiris et al. 2007; Vicente et al. 2010; Stage et al. 2015; Tigkas et al. 2020). In each procedure, the time series data of precipitation have important role (Svoboda and Fuchs 2016). In addition, accurate estimation and knowledge about the spatiotemporal variability of precipitation is the most important phase before drought monitoring and forecasting.

As, the precipitation and temperature are the primary important variables in meteorology, climatology and spatial statistics (Webste and Oliver 2007). Therefore, the precise estimation of precipitation data is unavoidable for accurate drought monitoring (Sun et al. 2018; Ali et al. 2019a, b; Jiang et al. 2020). In this study, we aimed to improve the accuracy of drought monitoring by enhancing precipitation time series data as the quality of meteorological data (i.e., preciseness, accurateness, and representativeness) has significant importance in inferencing natural hazards at the regional level. Therefore, this study used a combination of various auxiliary information-based sampling estimators to enhance the quality of drought indicators. In statistics and survey sampling, auxiliary information enhance the precision of an estimator and is widely used for improving the accuracy of estimated results (Cochran 1940). Since the temperature has a varying dependence structure with precipitation in various climatological regions. Therefore, this research provides a generalized compound framework for improving precipitation estimates. In this study, the compound framework is utilized for agricultural drought. However, expansion in computation can lead to the assessment of agriculture and hydrological drought. In previous research, Ali et al. (2019a, b) have

used minimum and maximum temperature as auxiliary information with total precipitation as a study variable for improving the precipitation records. Likewise, Jiang et al. (2020) have worked for regional drought monitoring.

The novelty of this research is to propose a compound framework of drought measure: The Seasonally Transient Weighted Multi-Scaler Standardized Index (STWMSDI). The framework of STWMSDI improves precipitation data by considering the correlation structures of observed precipitation with temperature. Under varying correlation structure, STWMSDI is based on ratio and product estimators. Hence, the proposed framework STWMSDI is more general than presented in Ali et al. (2019a, b).

2 The Proposed Framework—The Seasonally Transient Weighted Multi-Scaler Standardized Index (STWMSDI)

This section provides the outlines and methods used in the proposed framework for data improvement—STWMSDI. Figure 1 depicts the flowchart of the steps involved in the proposed framework. The following subsections provide the outline of STWMSDI.

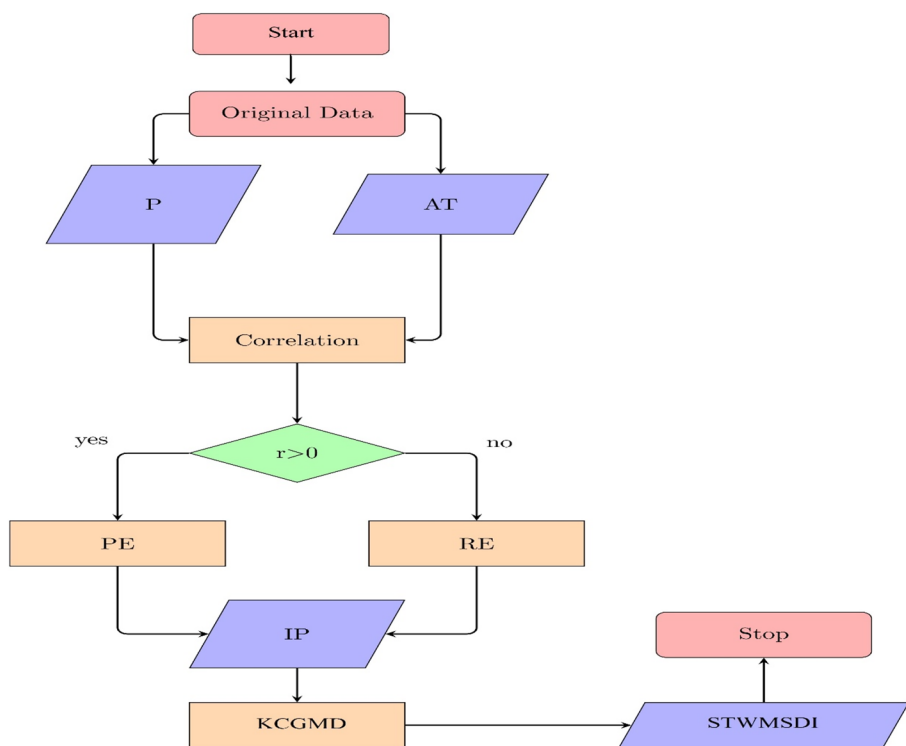


Fig. 1 Flowchart for the underlying methodology of proposed drought index: STWMSDI. Precipitation (P), Average Temperature (AT), Correlation coefficient (r), Product Estimator (PE), Ratio Estimator (RE), Improved Precipitation (IP), K-Component Gaussian Mixture Distribution (KCGMD), Seasonally Transient Weighted Multi-Scaler Standardized Index (STWMSDI)

2.1 Outlines of the proposed framework

To ease in understanding, we have provided the following eight steps that summarized the outlines and execution procedure of the proposed framework. The details on each step are as follows:

Step 1. Collection and screening of precipitation data

This step is straightforward. Here, we require total monthly precipitation and the monthly average temperature at certain meteorological stations.

Step 2. Segregation of temporal data

Although our proposal focused on the data improvement by using temperature as auxiliary information, literature shows that there is varying nature of the association of temperature with precipitation (Zhao et al. 1993). Further, this association also varies from region to region (Rebetez et al. 1996; Jaya et al. 2020). Therefore, to account for seasonality and make our proposal more general, this step suggests segregating time series data of precipitation and temperature based on the monthly indicator.

Step 3. Inspection of statistical relationship

After the data segregation, this step search association between precipitation and temperature data independently for each month. In the study, we suggest to employ the Pearson correlation (r) to quantify the relationship between study and auxiliary variables.

Step 4. Integration of sampling estimators

At a particular gauge station, the nature of the association (positive and negative significance relation) between precipitation and temperature will deploy the appropriate estimator for data improvement in each month, separately. In this step, we suggest preparing a list of candidate sampling estimators for data improvement under given auxiliary information.

Step 5. Data improvement

This step involved numerical exercise for data improvement by choosing an appropriate sampling estimator from the candidate list.

Step 6. The amalgamation of improved data

After data improvement, this step suggests an amalgamation of all the data in the standard time series model. By the assumption of mathematical statistics and sampling theory, the new time series data will be more representative than the traditional/usual observed data.

Step 7. Selection of appropriate probability model

Several studies have suggested probabilistic standardization for meteorological indicators. Therefore, appropriate univariate probability functions have been used in numerous articulated. However, experimental results show that the long time-series data have multiple locations or peaks. To increase more accuracy, we use the KCGMD as a novel approach. Furthermore, mathematical details on KCGMD have been provided in Sect. 2.3.

Step 8. Standardization

After fitting the probability model, this step suggests standardized and classify the improved data under the proposed combination of estimators. In this study, we have followed the standardization procedure provided in Vicente et al. (2010) and McKee et al. (1993). This procedure is quite common and has wide applications in hydrological research.

2.2 Auxiliary Information Based Sampling Estimators

The section will describe the rationale and deployment of statistical methods involved in the proposed framework.

To improve the estimate of any random phenomena, mathematical-statistical and sampling theory have provided a lot of estimators under certain situations. As the information about precipitation of gauge-based monitoring stations is based on single-point records, the surrounding environment near the station is not homogenous in most cases (Dutton et al. 2002). Furthermore, it is challenging to measure precipitation records in an optimized way (Parker et al. 2017). The recording of precipitation also depends on technology installation their maintenance, and financial resource. However, the temperature is globally representative, and its spatial distribution is mostly homogenous (Bräuning et al. 2004). Moreover, some studies have shown that there is a strong association between precipitation and temperature.

There is a positive correlation between precipitation and temperature (Madden et al. 1978). However, the correlation between precipitation and temperature is not always positive. Further, the correlation is varying spatially and temporally in different regions (Mondal et al. 2015) i.e. in a different month and season, the nature of correlation is varying. Zhao et al. (1993) have analyzed monthly total precipitation and temperature for 1000 metrological stations over eighty years across the united states. They have observed both negative and positive correlations in different months. Their study showed that summer precipitation and temperature are negatively correlated.

We are proposing a more general and state-of-the-art procedure for improving precipitation data under temperature as auxiliary information. In literature, several authors have provided various estimators under certain situations. Some of them are Vishwakarma et al. (2015); Verma et al. (2014); Solanki et al. (2013); Singh et al. (2012), etc. However, the regression, product, and ratio estimators are the most important and extensively used estimators. The integration of these estimators is based on the nature of the relationship between study and auxiliary variables.

In our proposed framework, instead of suggesting a single estimator as provide in Ali et al. (2019a, b), we are suggesting those estimators which are more appropriate under given constraints.

Hence, the proposed framework of using various estimators under various situations is theoretically sound because the correlation between temperature and precipitation varies with region and season. Consequently, this research employs ratio and product estimators to update monthly mean precipitation time series using mean monthly temperature as auxiliary information. The mathematical equations of the proposed estimators are expressed in the next subsection.

2.3 Data Improvement Under Multiple Auxiliary Estimators Settings

We have assessed that temperature has a varying correlation structure with precipitation in various regions and seasons from the literature. This study recommends average monthly temperature as auxiliary information for ratio and product to enhance the representativeness of precipitation data. The mathematical expression of these estimators for improving precipitation data under temperature as an auxiliary variable defines separately. Here, the expression of the ratio estimator is as follows:

$$LWM_{rat_i} = \frac{\bar{P}_{ij} \times \bar{T}_{ij}}{t_{ij}} \quad (1)$$

In above expression, LWM_{rat_i} is the locally monthly-wise improved mean precipitation records. $\bar{P}_{ij}$ is the moving mean of study variable for i_{th} period and j_{th} month, $\bar{T}_{ij}$ is the moving mean of auxiliary variable for i_{th} a period and j_{th} month, T_{ij} is the mean of auxiliary variable for i_{th} period and j_{th} month. Where, $i=3,6,12$ (time scale) and j representing months.

Similarly, the product estimator can be defined using temperature as an auxiliary variable as,

$$LWM_{Pr_i} = \frac{\bar{p}_{ij} \times \bar{t}_{ij}}{T_{ij}} \quad (2)$$

where LWM_{Pr_i} is the locally monthly-wise improved mean precipitation record. LWM_{Pr_i} is the locally monthly-wise improved mean precipitation record. $\bar{p}_{ij}$ is the moving mean of study variable for i_{th} period and j_{th} month, $\bar{t}_{ij}$ is the moving mean of auxiliary variable for i_{th} period and j_{th} month, T_{ij} is the mean of auxiliary variable for i_{th} period and j_{th} month. Where, $i=3,6,12$ (time scale) and $j=1, 2, 3, \dots, 12$ (Months).

Finally, we suggest combining the resultant data from the above two expressions so that the ratio estimator will be used if the correlation is negative. In the case of positive correlation, the product estimator should be employed. The following conditional equation can map the resultant combined time series:

$$CLWM_i = \begin{cases} LWM_{pr} & \text{if } r > 0 \\ LWM_{rat} & \text{if } r < 0 \end{cases} \quad (3)$$

$CLWM_i$ is a Combined Locally Weighted Monthly-Wise Estimates and is based on product and ratio estimators.

2.4 The Choice of a Probability Model

This study proposes a reliable and representative (STWMSDI) drought index based on the proposed $CLWM_i$ estimator of precipitation instead of using simple precipitation (p) time series directly. The proposed estimator is more representative of drought assessment because it accounts for the temperature that has directly correlated with precipitation. Replacing the sample record of precipitation (p) by $CLWM_i$ is logically and mathematically valid. Therefore, we have used $CLWM_i$ for obtaining the Standardized Drought index (SDI). The standardized procedure of SDI based on $CLWM_i$ is presented as follows.

One of the major aspects in the computation of multi-scaler drought index is the choice probability model on the time series data. In literature several fitting techniques are adopted for the estimation of SDI. In the beginning, gamma distribution was introduced as suitable destitution for SPI (Edwards et al. 1997). Guttman (1999) used Pearson Type III distribution for SDI standardization. Recently Ali et al. (2019a, b); Jiang et al. (2020) have fitted thirty-six propagate different distribution through r package (Spiess, 2014). However, more than one peak of data might be possible (multi-distribution); fitting a single distribution in such case can be poor fitting and can reduce the accuracy of drought monitoring. We used (KCGMD) as it is the best alternative for conventional fitting procedures.

KCGMD has several theoretical and computational advantages over the unimodal practice. It can fit all the possible distributions on a single iteration.

The functional form of KCGMD is as follows:

$$P(x) = \sum_{i=1}^K \alpha_i N(x|\mu_i, \sigma_i) \quad (4)$$

$$N(x|\mu_i, \sigma_i) = \frac{1}{\sigma_i \sqrt{2\pi}} \exp\left(\frac{-(x - \mu_i)^2}{2\sigma_i^2}\right) \quad (5)$$

$$\sum_{i=1}^K \alpha_i = 1 \quad (6)$$

where K represents the number of components, α_i is the mixture component weight of i_{th} component with the constraint $\sum_{i=1}^K \alpha_i = 1$, so that the total probability distribution normalizes to 1, μ_i , and σ_i are the mean and variance of i_{th} component. KCGMD has two types of parameters, the weight of mixture components and the mean and variance of each component.

In KCGMD, the selection of K is subjective to the nature of data. In this study, we have used 12 –component CGMD functions. As the data is collected under a 12-month structure, therefore, the choice of 12 components is rationally valid.

Following equation shows the combined form Cumulative Distribution Function (CDF) of 12-CGMD for improved data.

$$T(CLWMI) = F(CLWMI)_{\{1\}} + F(CLWMI)_{\{2\}} + \dots + F(CLWMI)_{\{12\}} \quad (7)$$

In the above equation, $T(CLWMI)$ is the combined CDF while $F(CLWMI)_{\{k\}}$ $\{k; 1, 2, 3, \dots, 12\}$ are the CDF of each component of Gaussian Distribution.

2.5 Standardization

Finally, the time series vector $T(CLWMI)$ is standardized using appropriate transformation method. Following the rationale of Wilhite (1996), we suggest to standardize the cumulative distribution function using the following set of equations provided by (Abramowitz et al. 1948). The appropriate transformation is carried out for both SPI and STWMSDI.

$$STWMSDI = -(z + \frac{c_0 + c_1 z + c_2 z^2}{1 + d_1 z + d_2 z^2 + d_3 z^3}) \quad (8)$$

For

$$z = \sqrt{\ln\left(\frac{1}{\{T(CLWMI)_{\{i\}}\}^2}\right)} \quad (9)$$

when,

$$0 \leq CLWMI_{\{i\}} \leq 0.5 \quad (10)$$

And,

$$STWMSDI = +(z - \frac{c + c_1z + c_2z^2}{1 + d_1z + d_2z^2 + d_3z^3}) \quad (11)$$

where,

$$z = \sqrt{\ln(\frac{1}{\{1 - CLWM_{\{i\}}\}^2})} \quad (12)$$

when,

$$0.5 \leq CLWM_{\{i\}} \leq 1 \quad (13)$$

$c_0 = 2.515517$, $c_1 = 0.802853$, $c_2 = 0.010328$, $d_1 = 1.432788$, $d_2 = 0.985269$, $d_3 = 0.001308$ are constants and $T(CLWM_{\{i\}})$ is the probability function.

3 Quality Measure and Comparative Assessment Indicator

In this paper, we compare our proposed index STWMSDI with the existing popular SPI drought indicator. Our comparative statistics include the Pearson product moment correlation coefficient (r). The correlation coefficient (r) is a common statistical tool used for the assessment, comparison, and the selection of drought indices (Vicente-Serrano et al. 2010; Leelaruban et al. 2017; Xia et al. 2020).

In this research, the assessment of the quality of the proposed index STWMSDI is performed with SPI with a three-month, six-months, and twelve-month-time scale, separately. In the context of SPI and STWMSDI, the general expression of r is given below:

$$r = \frac{\sum (SPI_i - \overline{SPI})(STWMSDI_i - \overline{STWMSDI})}{\sum (SPI_i - \overline{SPI})^2 (STWMSDI_i - \overline{STWMSDI})^2} \quad (14)$$

A positive but near to 1 value of r indicates a strong association between SPI and SWDP. In contrast, negative but near to -1 indicate a strong inverse relationship. Whereas, between 0.5 to -0.5 shows intermediate or weak positive or negative relation, accordingly.

4 Application

In application, ten representative meteorological stations namely Astore (35.36 N, 74.84E) Bunji (35.64 N, 74.63E), Chilas (35.43 N, 74.08E), Gilgit (35.92 N, 74.30E), Gupis (36.22 N, 73.44E), Skardu (35.30 N, 75.61E), Kotli (33.51 N, 73.90E), Muzaffarabad (34.35 N, 73.47E), Garhi Dupatta.

(73.61 N, 34.22E) and Chitral (35.83 N, 71.78E) of the northern area of Pakistan are covered (see Fig. 2). The selected meteorological stations are located on the second-highest plateau of the world has more than 20 peaks (above 6,100 m) including K2 (the second highest peak of the world. Therefore, the northern region has a significant impact on the whole hydrological system of Pakistan. In recent decades, Pakistan is facing significant deficient in rainfall due to worldwide global warming and climate change. Consequently, drought has become one of the major problems for agriculture, energy, human life, and the ecosystem. In recent research, several authors have

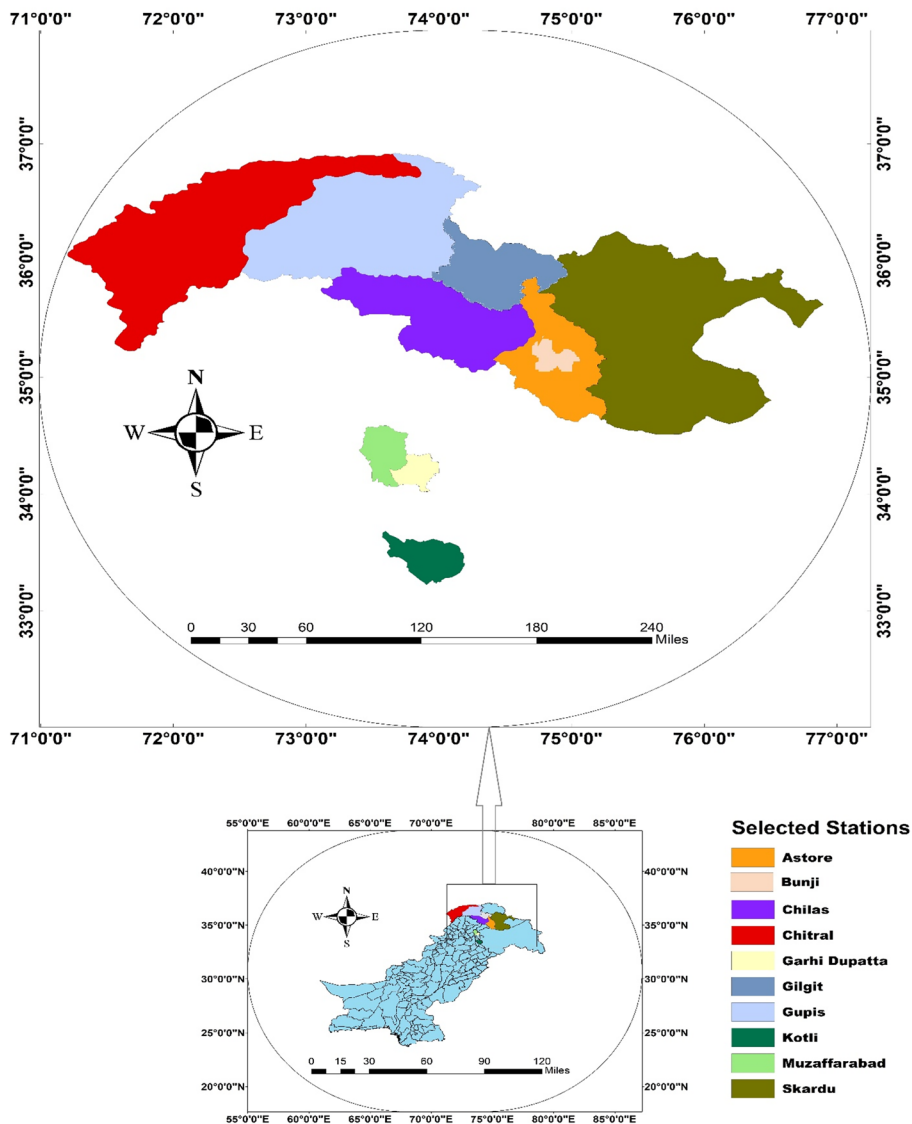


Fig. 2 Geographical distribution of the selected meteorological stations

quantified drought and provided various methods for drought monitoring for the northern area of Pakistan (Ali et al. 2017). For example, (Ali et al. 2019a, b) quantify the drought classes and have provided a Probabilistic Weighted Joint Aggregative scheme for drought monitoring in the said region.

For this research, long time series data ranged from 1970–2017 of monthly accumulated precipitation (p), an average of minimum and maximum temperature (T_{min} , T_{max}) is collected from Pakistan Meteorological Department. Table 1 shows the summary statistics of precipitation and temperature of five meteorological stations.

Table 1 Summary statistics of the some of the selected stations

		Stations									
		Astore		Bunji		Chillas		Gilgit		Gupis	
		Prec	Temp	Prec	Temp	Prec	Temp	Prec	Temp	Prec	Temp
Jan	Avg	37.55	-5.64	6.13	5.16	10.7	7.63	3.66	2.24	7.6	1.86
	Std	37.88	2.58	8.02	1.61	16	1.51	6.31	1.68	11.6	2.12
	Kurt	1.61	-0.56	6.35	-0.86	5.26	-0.74	10.14	-0.46	3.57	-0.77
Feb	Avg	45.84	-2.98	9.79	9.28	16.1	11.11	7.15	6.82	13.11	5.38
	Std	35.75	2.43	14.6	1.75	15.4	1.82	8.9	1.81	26.98	2.32
	Kurt	0.29	1.03	8.94	1.03	4.96	0.44	4.36	0.79	23.66	0.68
Mar	Avg	70.32	3.74	12.1	17.02	28.1	18.59	11.31	14.68	16.82	13.51
	Std	44.32	2.27	12	2.21	26.5	2.26	11.03	1.8	17.95	2.75
	Kurt	1.23	-0.06	7.42	-0.69	1.79	-0.06	2.58	-0.67	1.45	-0.55
Apr	Avg	77.9	11.8	25.6	24.5	35.6	27.22	24.84	21.55	43.09	22.6
	Std	52.25	1.92	25.7	1.75	33.5	1.96	28.19	1.36	75.04	2.21
	Kurt	0.15	-0.18	1.5	-0.24	1.39	-0.37	3.76	0.24	11.14	-0.53
May	Avg	61.15	17.8	26.3	29.66	30.9	34.67	25.45	26.5	27.08	29.26
	Std	51.1	2.3	26.9	2.83	35.8	3.37	24.06	2.08	30.54	3.43
	Kurt	-0.19	-0.27	2.72	-0.13	3.94	-0.23	-0.1	0.15	0.81	0.08
Jun	Avg	25.95	23.7	11	35.86	10.9	42.59	10.91	32.03	18.74	36.34
	Std	20.74	2.13	10.3	2.92	13.9	2.78	9.19	2.11	28.1	2.78
	Kurt	5.36	0.7	0.16	0.56	5.6	0.32	4.83	1.47	5.19	0.14
Jul	Av	25.7	28	17.5	40.3	12.6	46.59	14.95	36.06	16.18	40.5
	Std	19.94	1.94	16.1	2.51	14	2.18	12.66	2.08	18.17	2.43
	Kurt	2.28	0.05	2.65	-0.02	5.35	-0.52	1.49	0.21	4.61	-0.03
Aug	Avg	26.55	27	22.8	39.19	17.6	45.4	17.21	34.93	26.4	38.93
	Std	22.13	1.64	23.9	2.92	18.5	2.26	18.6	1.8	29.05	2.66
	Kurt	5.81	0.15	2.61	0.1	7.77	0.53	4.46	0.62	8.29	-0.48
Sep	Avg	27.19	21.8	15.2	32.4	10	39.55	11.25	28.25	13.55	32.75
	Std	33.06	1.65	19	2.4	20.8	2.02	12.65	1.52	21.94	1.89
	Kurt	8.31	-0.56	2.31	0.93	24	-0.81	6.25	0.88	12.93	-0.17
Oct	Avg	21.04	13.5	6.91	23.43	6.7	29.21	6.35	19.28	8.82	23.83
	Std	29.78	1.52	19.2	1.88	22	1.97	16.31	1.35	17.17	2.13
	Kurt	15.1	0.21	35.4	-0.48	36	-0.08	30.21	-0.85	19.67	-0.63
Nov	Avg	19.48	5.53	3.26	14.07	6.18	17.99	3.31	9.88	2.8	14.75
	Std	25.7	1.63	8.53	1.5	12.8	1.22	5.82	1.22	6.51	1.53
	Kurt	3.33	-0.67	18	2.06	6.25	-0.41	6.39	0.8	8.14	-0.37
Dec	Avg	29.79	-1.53	5.51	7.25	10.6	9.89	5.12	3.72	4.77	5.2
	Std	38.71	2.06	9.91	1.46	18.8	1.52	8.66	1.73	9.67	1.76
	Kurt	5.1	-0.35	13.9	0.15	5.51	-0.78	5.21	0.12	12.65	-0.26

5 Result and Discussion

In this research, we have used monthly time series data ranging from 1970 to 2016 of average precipitation and mean temperature for ten meteorological stations located in the

northern area of Pakistan. To compare and validate the proposed index, we have computed SPI-3, SPI-6, and SPI-12 under the same computational settings which are used for the proposed index as well.

In Table 2, correlation and its related p-values between precipitation and mean temperature at various months in all stations are presented. To improve precipitation data, we employed mean temperature as auxiliary information (see the flowchart in Fig. 1). Accordingly, the nature of the correlation values shows the recommendation of product and ratio estimators for precipitation data improvement. Under the setting of our proposal, the ratio estimator (see Eq. 3) is used to improve precipitation data in a specific month where the correlation between precipitation (study variable) and temperature (as an auxiliary variable) is positive. Accordingly, for negative correlation, a product estimator is employed (see Eq. 4). We have observed that, in most of the months in all stations, there is a negative correlation between precipitation and temperature. However, in some stations and months as well, a positive correlation is reported. Therefore, our proposal related to data improvement requires a varying selection of estimators in varying months and stations as well.

To check and validate the results of our model, the deviation between the improved and simple precipitation data is reported using density plots.

Figure 3 shows the deviation in the distributions of precipitation data of simple and improved records. Outcomes show substantial deviation between simple and improved records. Hence, the use of temperature as auxiliary information is rationally valid. Further, for accurate analysis, reporting drought events, and climatological characterization of a particular region in a particular month, advance sample techniques and estimators strengthen the quality and reliability of results. In addition, the proposed framework provides a solid way to fulfil the requirement of a multivariate drought index (Vicente et al. 2010; Hao et al. 2013) for reliable drought analysis and drought quantification.

Table 3 shows the Bayesian information criteria (BIC) values of the 12-Component Gaussian Mixture Distribution for Simple and Improved data for all the selected stations at various time scales. In this probabilistic modeling, a substantial deviation between the distribution characteristics of in usual and improved data has also been observed.

For instance, BIC values in the meteorological stations, Astore, Bunji and Chilas at three-month time scales of simple precipitation records are -4753.49, -3749.12, and -3882.64, respectively. While, in improved records, the BIC values are -4837.08, -3559.62, and -3841.91, accordingly. Similarly, the significant deviation among other stations and time scale can be observed from Table 3.

To assess the competency of our proposed framework, we have standardized the fitted CDF of 12 component Gaussian distribution on both types of precipitation records.

Further, the consistency between both time-series data of standardized values is observed using Pearson correlation measure.

In the Astore station, the correlation between the proposed and existing methods are 0.9121, 0.9215 and 0.9285 for time-scale-3, time-6 and time-scale-12, respectively. In Skardu station, the correlation for scale-6 and scale-12 is 0.8664 and 0.8387 respectively. Overall, there is a significant high and positive correlation between the two methods (see Table 4). This indicates that STWSDI is a good candidate drought indicator that incorporates auxiliary information based improved precipitation records for drought analysis.

Figure 4 present the density plots of 12-components Gaussian mixture distribution parallel with their Q-Q plots for simple and improved data for two stations. These Q-Q plots in all stations show that the distribution is the best fit on the precipitation time series. In another word the densities cover maximum data values. We have observed that the Gaussian mixture distribution is a more appropriate model instead of fitting a single model.

Table 2 Correlation structure between precipitation and mean temperature at various months

Months		Astore	Bunji	Chilas	Gilgit	Gupis	Skardu	Kotli	Muzaffarabad	Gari-Dupatta	Chitral
Jan	(r)	0.0850	0.1230	-0.2560	0.1360	-0.1790	-0.0510	-0.0450	-0.2570	-0.0230	-0.3280
	P-values	0.5710	0.4090	0.0820	0.3630	0.2290	0.7320	0.7660	0.0810	0.8790	0.0240
Feb	(r)	0.0510	-0.0080	-0.1640	0.1440	-0.0760	-0.1980	-0.2270	-0.2970	-0.2300	-0.3960
	P-values	0.7330	0.9590	0.2700	0.3350	0.6130	0.1820	0.1250	0.0420	0.1190	0.0060
Mar	(r)	-0.4600	-0.3480	-0.5280	-0.2190	-0.4150	-0.5190	-0.6430	-0.6970	-0.7510	-0.6890
	P-values	0.0010	0.0160	0.0000	0.1400	0.0040	0.0000	0.0000	0.0000	0.0000	0.0000
Apr	(r)	-0.2140	-0.3400	-0.3170	-0.1350	-0.3550	-0.4450	-0.5880	-0.5540	-0.4050	-0.2920
	P-values	0.1480	0.0190	0.0300	0.3650	0.0140	0.0020	0.0000	0.0000	0.0050	0.0460
May	(r)	-0.7600	-0.5890	-0.7030	-0.4790	-0.5970	-0.5520	-0.4260	-0.5620	-0.3940	-0.6160
	P-values	0.0000	0.0000	0.0000	0.0010	0.0000	0.0000	0.0030	0.0000	0.0060	0.0000
June	(r)	-0.3460	-0.2830	-0.3990	-0.2690	-0.4690	-0.2720	-0.1230	0.0460	-0.0010	-0.5850
	P-values	0.0170	0.0540	0.0050	0.0670	0.0010	0.0650	0.4120	0.7570	0.9930	0.0000
July	(r)	-0.2560	-0.2830	-0.4570	-0.2350	-0.3450	-0.2660	-0.2800	-0.0930	-0.3020	-0.4070
	P-values	0.0830	0.0540	0.0010	0.1110	0.0180	0.0710	0.0560	0.5330	0.0390	0.0050
Aug	(r)	-0.2120	-0.3760	-0.3780	-0.3750	-0.4070	-0.2790	-0.2920	-0.1540	-0.0410	-0.2910
	P-values	0.1520	0.0090	0.0090	0.0090	0.0040	0.0570	0.0470	0.3010	0.7860	0.0470
Sep	(r)	-0.3240	-0.1790	-0.3950	0.0060	-0.3940	-0.1930	-0.2670	-0.2010	-0.1980	-0.0660
	P-values	0.0260	0.2300	0.0060	0.9710	0.0060	0.1930	0.0690	0.1760	0.1810	0.6580
Oct	(r)	-0.3820	-0.2090	-0.3920	-0.0230	-0.4390	-0.0250	-0.2180	-0.2400	-0.1460	-0.6040
	P-values	0.0080	0.1590	0.0060	0.8810	0.0020	0.8700	0.1410	0.1040	0.3290	0.0000
Nov	(r)	-0.4130	-0.2760	-0.3080	0.0020	-0.4040	0.1080	0.2100	-0.0670	0.1400	-0.2250
	P-values	0.0040	0.0600	0.0350	0.9890	0.0050	0.4710	0.1560	0.6560	0.3490	0.1290
Dec	(r)	-0.4750	-0.0530	-0.1370	0.3030	-0.1780	-0.1160	-0.0280	-0.0940	0.0260	-0.1730
	P-values	0.0010	0.7240	0.3600	0.0380	0.2330	0.4390	0.8490	0.5290	0.8610	0.2440

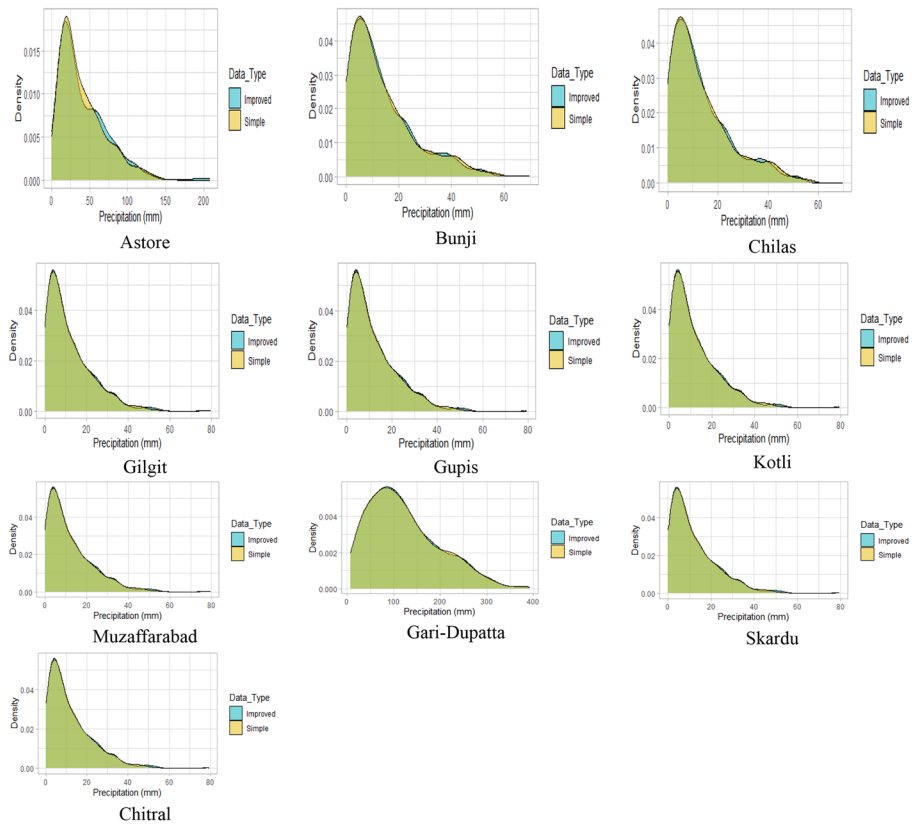


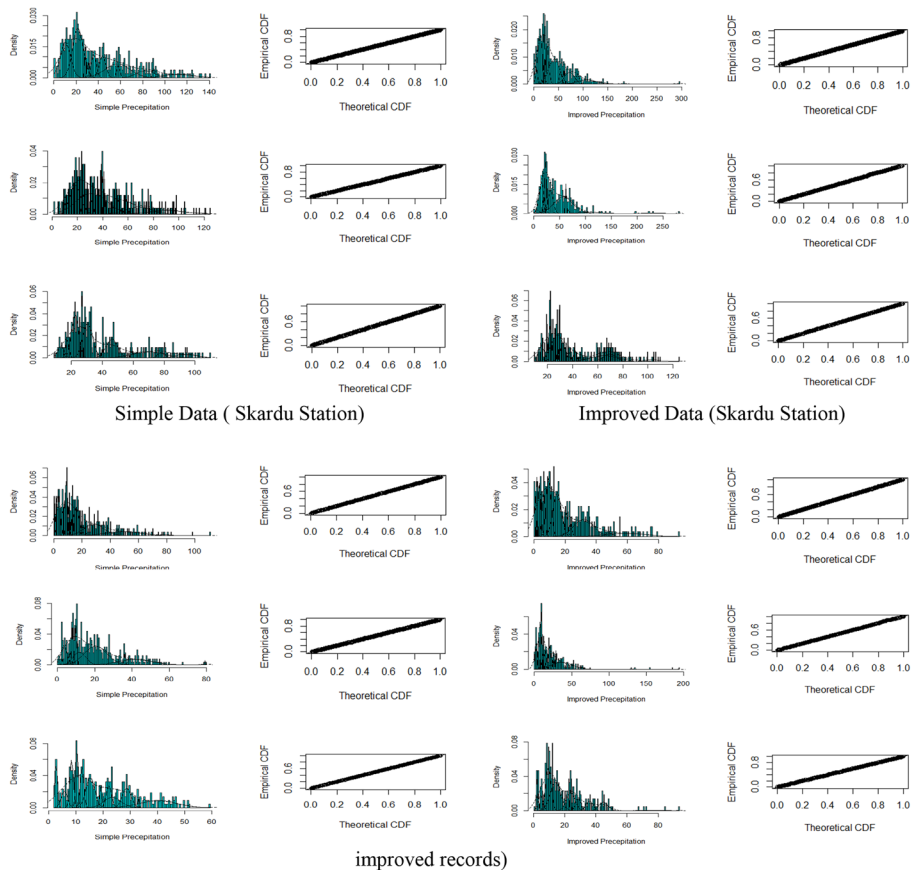
Fig. 3 Density plots showing deviation in simple and improved precipitation in the selected stations

Table 3 BIC values of the 12-CGMM

Stations	Time Scales					
	Three-Month		Six-Months		Twelve-Months	
	Simple	Improved	Simple	Improved	Simple	Improved
Astore	-4753.49	-4837.08	-4278.77	-4345.39	-3468.39	-3455.34
Bunji	-3749.12	-3559.62	-3723.21	-3314.25	-2724.41	-2736.01
Chilas	-3882.64	-3841.91	-3523.03	3527.89	-2859.54	-2876.9
Gilgit	-3563.84	-3564.99	-3232.19	-3227.93	-2608.39	-2618.90
Gupis	-3847.66	-4352.59	-3591.91	-3569.06	-3042.41	-3033.05
Skardu	-4109.28	-4059.36	-3684.98	-3767.38	-2933.65	-3025.72
Kotli	-5818.88	-5827.45	-5288.17	-5292.05	-4298.55	-4308.53
Muzaffarabad	-5978.97	-5968.03	-5310.87	-5445.10	-4488.36	-4493.39
Gari-Dupatta	-5892.17	-5888.51	-5389.83	-5383.18	-4474.74	-4675.68
Chitral	-4793.86	-4693.27	-4442.17	-4409.57	3738.81	-3736.78

Table 4 Correlation between SPI and STWMSDI in various time scales at the selected stations

Stations	Time Scales		
	Three-Month	Six-Months	Twelve-Months
Astore	0.9121	0.9215	0.9285
Bunji	0.9959	0.9965	0.9972
Chilas	0.9972	0.9943	0.9984
Gilgit	0.9842	0.991	0.9973
Gupis	0.9583	0.9151	0.9897
Skardu	0.9773	0.8664	0.8387
Kotli	0.9985	0.9986	0.9989
Muzaffar-Abad	0.9975	0.9356	0.9984
Gari-Dupatta	0.9954	0.993	0.9949
Chitral	0.9842	0.9895	0.9962

**Fig. 4** Probability density and Q-Q plots 12-CGMD at various time scales in Astore and Skardu Stations (Left sides plots are based on simple records, Right sides graphs are based improved records)

Additionally, this study includes a temporal behavior of STWSDI and SPI for all station and for all time scale. For example, the pattern of temporal values of the proposed index STWSDI is very homogenous to those which have been estimated using simple precipitation records at 3-month time scale (see Fig. 5). Accordingly, the temporal behavior of STWSDI and SPI have been assessed. Here, we conclude that there is no contradiction between both records in drought characterization. The strong consistency and behavior of temporal plots demonstrate that STWSDI is a good indicator for monitoring and forecasting drought. Though some deviations are observed at Gupis station, it might be because of an outlier. Since the outlier is beyond the coverage of optimal probability distributions. However, in the rest of the stations, both of the series have homogenous temporal patterns.

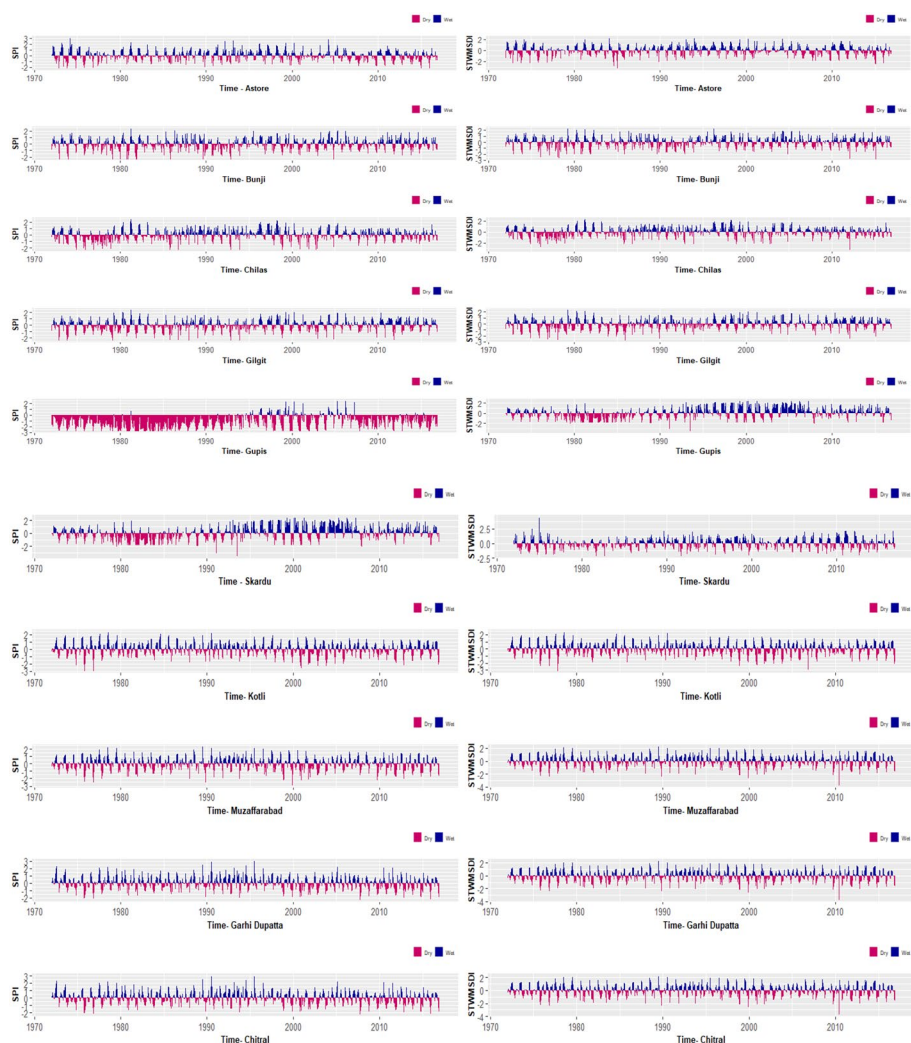


Fig. 5 Time series of SPI (Left side) and STWSDI (Right side) at 3-month time scale

6 Concluding Remarks

Drought has a significantly bad impact on society, agriculture, and the ecosystem. For drought mitigation policies, accurate drought monitoring has substantial importance in hydrological and meteorological research. In this paper, an attempt has been made to get more reliable and accurate results related to continuous drought monitoring. Here, we have introduced a novel framework—the Seasonally Transient Weighted Multi-Scaler Standardized Index (STWMSDI). Outcomes of this research have proven the acceptance of the proposed framework for enhancing the accuracy and reliability of precipitation data before its further uses. The key concluding remarks of this paper are:

- 1) The role of the auxiliary variable is very important for the improvement of the study variables.
- 2) There is a varying spatio-temporal correlation behavior of temperature with precipitation.
- 3) The proposed framework of STWMSDI enhances the accuracy and reliability of precipitation data.
- 4) The application of 12-CGMM in the computation of drought indicators is an excellent fitting technique.
- 5) To enhance precipitation data quality, auxiliary information strengthens the drought monitoring system.

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Data Availability The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Competing Interests The authors declare no competing interests.

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